

FCReL at Powertech Labs

12388 88 Avenue
Surrey, BC, V3W 7V8
November 30, 2025

Dr. John Shen, Director

School of Mechatronic Systems Engineering
Simon Fraser University
Surrey, BC, V5A 1S6

Dear Dr. Shen,

The enclosed report, titled “Assembly and Testing of a 5 cm² Tandem 50 Research Single Cell on a Greenlight G40 Test Station”, was prepared as a requirement for my second co-op work term in the Fall 2025 semester. The purpose of this report is to document the practical procedures, challenges, and outcomes of assembling a 5 cm² research single cell in Tandem 50 hardware, integrating it with the Greenlight Innovation G40 test station, and verifying safe and leak-tight operation.

During this work term at FCReL Powertech Labs in Surrey, British Columbia, I worked as a Research Assistant under the supervision of Dr. Erigene Bakangura, Head of Lab Operations. My responsibilities focused on hands-on tasks: stacking and aligning the membrane electrode assembly (MEA) in the Tandem 50 fixture, configuring the G40 gas arms, thermocouples and heaters, performing nitrogen and hydrogen leak checks, and running representative test scripts to confirm stable operation.

This work was conducted on the traditional, ancestral, and unceded territories of the q̓íc̓əy̓ (Katzie), q̓ʷa:n̓l̓ʼən̓ (Kwantlen), sc̓əwaθən məsteyəx̓w (Tsawwassen), and səmyámə (Semiahmoo) First Nations.

I hereby confirm that I have received no further help in writing this report other than what is stated above and that this report has not been previously submitted for academic credit at Simon Fraser University or any other institution.

Sincerely,



Thanh Nam Tran

Student ID: 301540228

Assembly and Testing of a 5 cm² Tandem 50 Research Single Cell on a Greenlight G40 Test Station

Co-op Technical Report – Work Term 2

Prepared for:

Dr. John Shen

School of Mechatronic Systems Engineering

Simon Fraser University

Prepared by:

Thanh Nam Tran

Student ID: 301540228

Employer:

Fuel Cell Research Lab at Powertech Labs Inc. , Surrey, BC

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Abstract

This report describes the hands-on assembly and testing of a 5 cm² proton exchange membrane (PEM) research single cell using Tandem 50 hardware on a Greenlight Innovation G40 test station at FCREL Powertech Labs. Building on the first co-op term, which involved assembling and operating a 0.39 cm² small-scale cell on a G20 station, the focus of this term was to develop a clear, repeatable procedure for assembling the larger Tandem 50 single cell, integrating it with the G40 platform, and verifying safe and leak-tight operation.

The membrane electrode assembly (MEA) was stacked using cathode and anode endplates, gold-coated current collectors, reaction distribution plates, and a polycarbonate spacer. Particular attention was given to plate orientation, MEA centring, uniform compression, and minimizing the risk of mechanical damage during stacking. After assembly, the cell was installed on the G40 test station, connected to heated gas arms, thermocouples, heater cables, and a pressure line, and subjected to nitrogen leak testing, hydrogen leak sniffing, and an electrochemical leak detection check.

The leak testing confirmed that the assembled cell was properly sealed at beginning of life, and representative open-circuit voltage (OCV) and polarization measurements verified stable operation under hydrogen/air conditions. The main outcome of this work is a detailed, step-by-step procedure for Tandem 50 MEA assembly, G40 integration, leak testing, and safe disassembly—providing a practical reference for future FCREL students and supporting the lab's ongoing fuel-cell durability studies.

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Background

What is a PEM Fuel Cell?

A proton exchange membrane (PEM) fuel cell converts the chemical energy of hydrogen and oxygen directly into electricity, heat, and water through electrochemical reactions at low operating temperature. Hydrogen is fed to the anode, where it splits into protons and electrons. The protons cross the polymer membrane while the electrons pass through an external circuit, providing useful electrical power. At the cathode, oxygen reacts with the protons and electrons to form water.

Because PEM fuel cells can operate quietly with high power density and only water as an exhaust product, they are widely investigated for applications such as transportation, backup power, and distributed generation.

What is the G40 Test Station and Tandem 50 Fixture?

The Greenlight Innovation G40 is a research-grade fuel cell test station capable of precisely controlling single-cell operating conditions such as gas flow rates, pressure, relative humidity, and temperature. It includes automated purges, interlocks, and data acquisition through the Emerald FCATS software. The G40 shares many features with the smaller G20 stations but is designed for higher flows and larger test fixtures.

The Tandem 50 hardware is a 5 cm² research single-cell fixture used at FCReL. It consists of cathode and anode endplates with integrated heaters, gold-coated current collectors, reaction distribution plates, a polycarbonate spacer defining the active area, and an external air-compressor plate that applies uniform clamping pressure. This fixture allows MEAs to be tested under realistic compression and flow-field conditions while still being accessible for assembly and inspection.

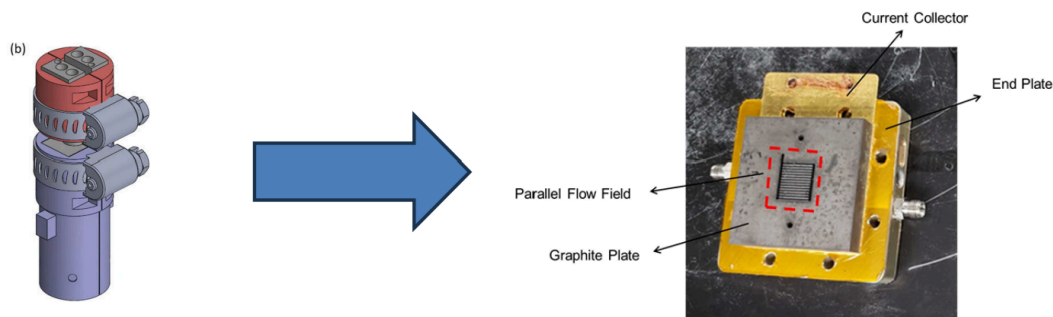


Figure 1 – Transition from the previous small-scale XCT-compatible single cell (left) to the 5 cm² Tandem 50 research single cell used in this work (right).

Glossary of Common Terms

- **PEM** – Proton Exchange Membrane
- **MEA** – Membrane Electrode Assembly
- **GDL** – Gas Diffusion Layer
- **PFSA** – Perfluorosulfonic Acid ionomer
- **OCV** – Open Circuit Voltage
- **ELDT** – Electrochemical Leak Detection Test
- **DUT** – Device Under Test (the fuel cell)

1. Introduction

1.1 Background and Context

FCReL Powertech Labs operates several Greenlight fuel cell test stations that support research on single-cell PEM fuel cell performance and durability. In the first co-op work term, a 0.39 cm² small-scale single cell was assembled and tested on a Greenlight G20 station, providing practical experience with MEA handling, leak checking, and basic operation of a PEM fuel cell.

For this second work term, the same workflow was carried out using a larger 5 cm² research single cell in the Tandem 50 fixture on a Greenlight G40 station. The purpose was to apply the skills learned previously to a different hardware configuration and to gain hands-on experience assembling the larger fixture, integrating it with the G40 station, and safely bringing the cell into operation under hydrogen and air.

1.2 Problem Statement

Scaling the assembly process from the previously tested small-scale hardware to the larger 5 cm² Tandem 50 fixture presented new integration challenges. The specific problem was to validate that the same MEA type could be successfully implemented in the Tandem 50 hardware and operated on the high-flow Greenlight G40 station. This required establishing a verified hands-on procedure to stack the fixture, confirm gas-tight integrity with nitrogen and hydrogen, and

achieve stable electrochemical performance without the mechanical failures or leaks often associated with larger compression areas.

1.3 Objectives

The objectives of this work term were to:

- Assemble a 5 cm² research single cell in the Tandem 50 fixture using a clear and repeatable stacking procedure.
- Integrate the assembled cell with the G40 test station, including gas arms, thermocouples, heaters, and pressure line, while maintaining mechanical stability and good thermal insulation.
- Perform nitrogen and hydrogen leak checks, including an electrochemical leak detection test, and record acceptable leak-test results.
- Acquire representative OCV and polarization data to confirm that the assembled cell can operate stably under hydrogen/air conditions.
- Summarize the assembly, setup, leak testing, and shutdown steps in a form that can be reused in future FCREL experiments.

2. Methodology

2.1 Tools, Equipment, and Materials

The following tools and equipment were prepared prior to assembly:

- Tandem 50 single-cell hardware:
 - Cathode and anode endplates (cathode endplate is the heating plate),
 - Gold-coated current collector plate,
 - Two reaction distribution plates (“green” and “black” sides),
 - Polycarbonate spacer plate,
 - External air-compressor plate,

- Tie rods, nuts, and alignment rods.
- Circular MEA and GDLs sized for a 4 cm diameter active region.
- Torque wrench and appropriate hex keys.
- Isopropanol (IPA) and lint-free wipes for cleaning plate surfaces.
- Kapton tape and insulation material (white cloth sleeves and tape) for thermal management.
- Greenlight G40 fuel cell test station with Emerald FCATS control software.
- Nitrogen and hydrogen gas supplies with dual-stage regulators and appropriate safety hardware.
- Leak detection fluid (soap solution) and a hydrogen gas sniffer.

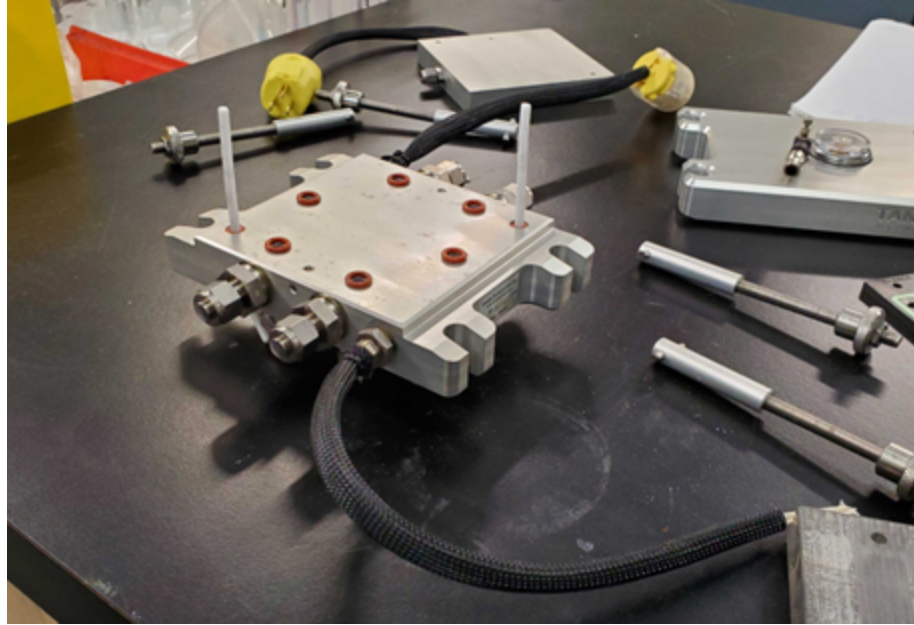
A clean, uncluttered work area was used to avoid contamination of the MEA and to make it easier to maintain the correct orientation of all components.

2.2 MEA Stacking in the Tandem 50 Fixture

The MEA and plates were stacked following the internal G40 setup notes, with particular attention to orientation and uniform pressure distribution.

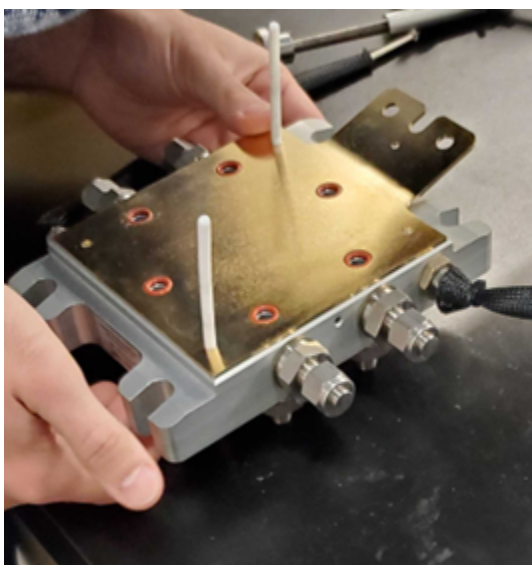
1. Cathode endplate orientation

- The cathode endplate (heating plate) was placed flat on the bench with the label facing up and used as the reference orientation for all subsequent components.
- The two white alignment rods in opposite corners were checked to ensure they were clean and straight.



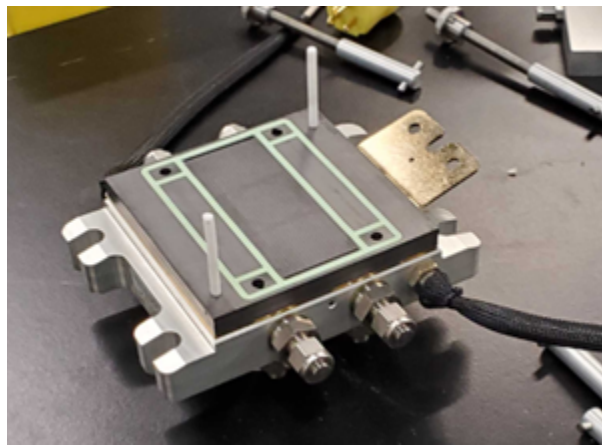
2. Gold-coated current collector plate

- The gold-coated plate was slid onto the two white rods at the designated corners.
- The “red circle dots” on the gold plate were aligned with those on the cathode endplate.
- The portion of the gold plate that sticks out (current tab) was oriented so it pointed upwards on the same side as the labels, ensuring later access for electrical connection.



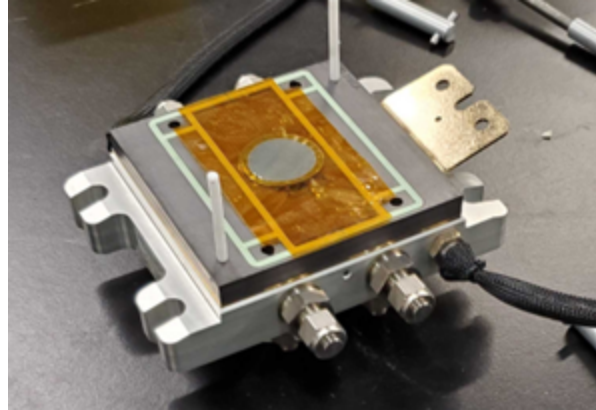
3. First reaction distribution plate

- One reaction distribution plate was placed onto the rods.
- The numbering engraved in the top-left corner and the “green” side were oriented so that the number was on the same side as the cathode endplate label.
- For this build, the “green” side was oriented facing up.



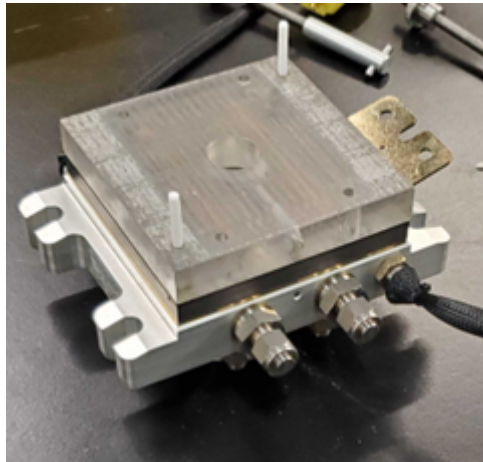
4. MEA placement

- The MEA was carefully placed on the reaction distribution plate.
- The edges of the membrane were visually centred so that the distance to the circular window in all directions was approximately equal. This step is important to avoid uneven compression and premature failure due to stress concentration at the edges.
- The shiny membrane side was facing upward.



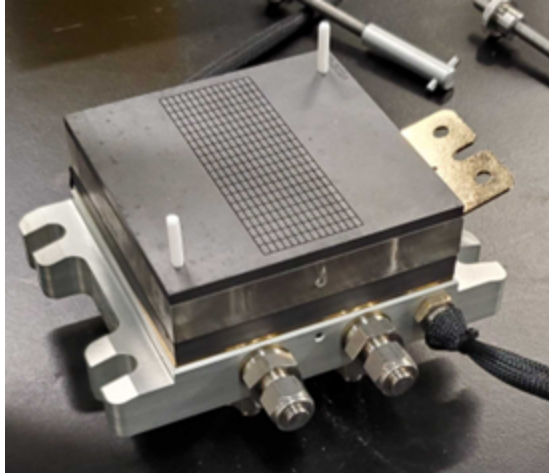
5. Polycarbonate spacer plate

- Starting from this step, extra care was taken not to disturb the MEA position.
- The polycarbonate spacer plate was lowered slowly over the rods and placed on the MEA, making sure the cut-out window was centred over the active area and the orientation matched the reference image in the notes.



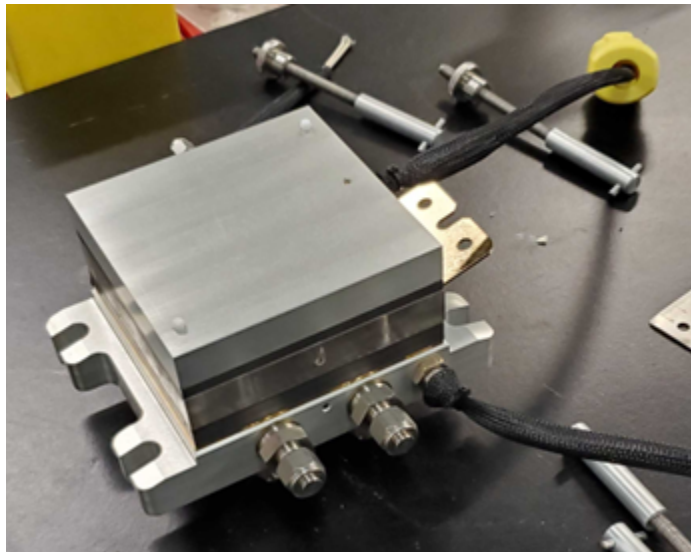
6. Second reaction distribution plate

- The second reaction distribution plate was installed with the “black” side facing the spacer and the engraved numbering again aligned with the labels.
- This ensured consistent orientation between the two plates.



7. Anode endplate

- The anode endplate was added on top, with its label on the same side as the cathode label.
- Alignment of all labels made it easier to identify anode and cathode directions after assembly.

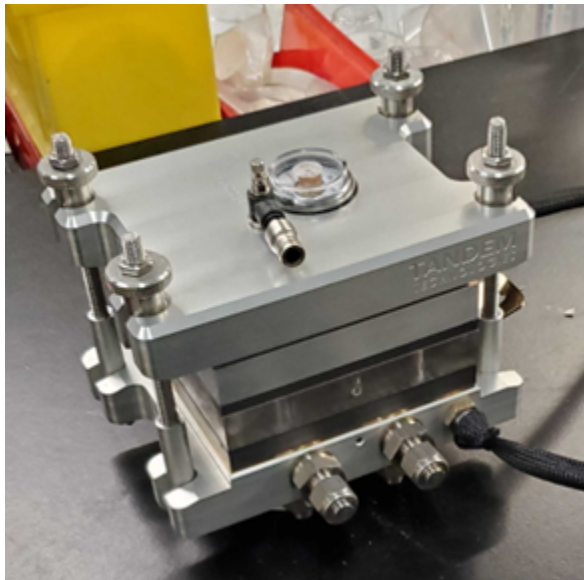


8. Air-compressor plate

- The external air-compressor plate was placed on top of the stack in the orientation specified in the notes, allowing it to apply uniform clamping pressure when connected to the pressure line.

9. Tightening sequence

- Nuts on the tie rods were tightened gradually and evenly.
- A diagonal tightening pattern was used: each nut was turned roughly two turns, then the opposite corner nut was tightened, and so on.
- This incremental pattern was repeated until the stack was snug and no plate gaps were visible.
- The goal was to distribute stress evenly across the MEA and avoid bending the plates, which could lead to non-uniform contact pressures.



Throughout stacking, the assembly was visually inspected from all sides to confirm that the plates remained parallel and that the MEA had not shifted.

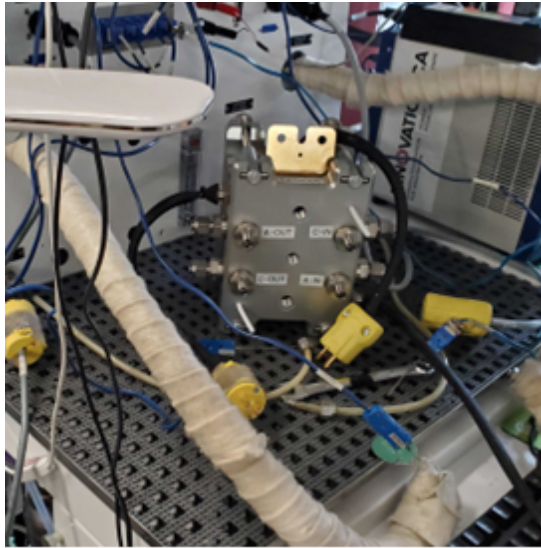
2.3 G40 Test Station Setup with Tandem 50

After stacking, the Tandem 50 assembly was transferred to the G40 test station and connected to the instrumented gas arms and wiring.

1. Placement on the station

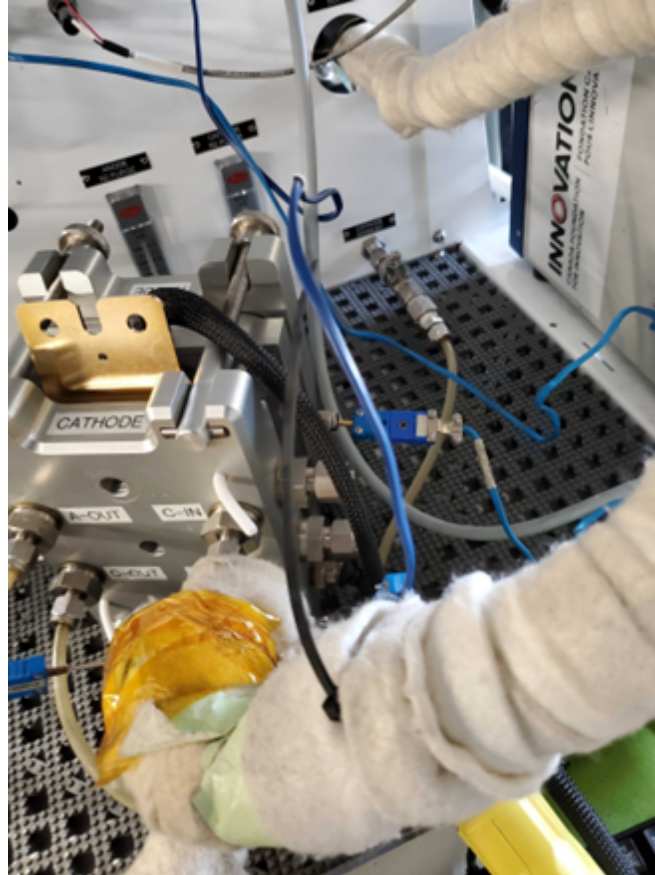
- The stack was placed on the G40 platform so that the labelled side faced slightly away from the station window.

- This small angle helps reduce bending loads from the flexible gas arms.



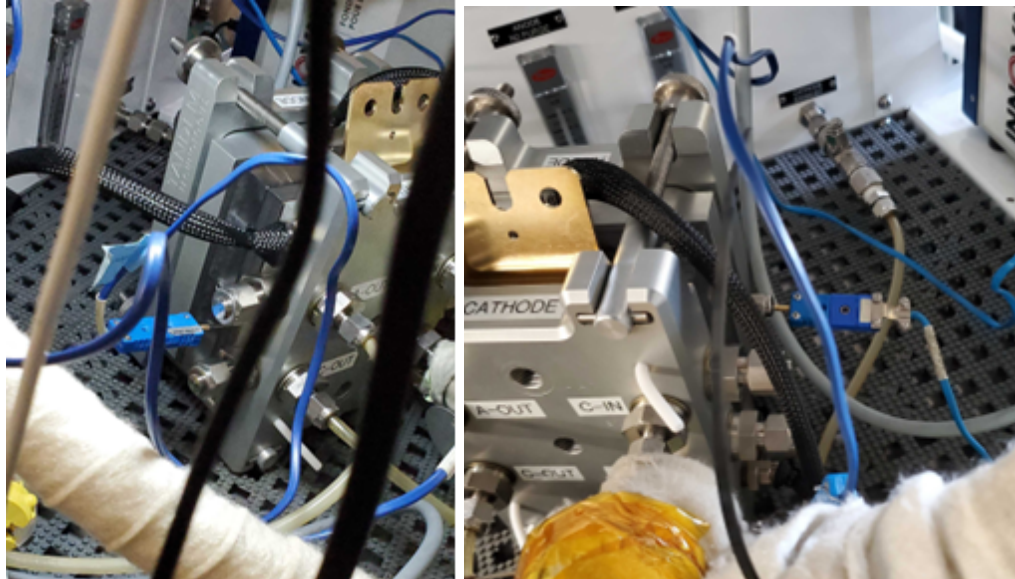
2. Connecting gas arms

- The four G40 gas arms were connected to the labelled fittings on the air-compressor plate:
 - C-IN: cathode supply (right side),
 - C-OUT: cathode return (right side),
 - A-IN: anode supply (left side),
 - A-OUT: anode return (left side).
- After connection, the mechanical stability of the stack was checked. If any arm caused visible tilting, its routing was adjusted.



3. Thermocouples

- Thermocouples were threaded into the ports on the endplates: left-side for anode, right-side for cathode, following Greenlight's convention.
- The thermocouple wires were routed away from the hot surfaces to reduce heat exposure and avoid direct contact with endplates.



4. Heaters and pressure line

- Heater cables (yellow plugs) from the G40 station were connected to the heater sockets on the air-compressor plate.
- The blue transparent tube used as the pressure nozzle was connected to the air-compressor plate to allow the station's air line to apply clamping pressure to the cell.

5. Insulation and wiring management

- The gas arms already had white insulation wrapped around them. Any areas where red insulation was visible were covered with extra insulation or tape to minimize heat loss.
- Electrical wires were tied or routed so they would not rest on hot endplates or be pinched by moving parts.

6. Software checks

- In the FCATS software, the correct device configuration was selected and channel assignments for thermocouples and heaters were verified.
- The safety interlocks and emergency stop buttons were tested before introducing gas.

2.4 Leak Detection and Safety Checks

Safe operation required a sequence of leak checks with nitrogen and hydrogen before moving to full fuel cell testing.

1. Nitrogen purge and external leak check

- With hydrogen valves closed, nitrogen was supplied to both anode and cathode using the Auxiliary purge controls in the G40 software.
- Flow was increased gradually while monitoring pressure and checking for anomalies.
- Leak detection fluid was applied to all external connections: regulators, quick connects, and fittings on the Tandem 50 hardware. Any bubbling would indicate a leak and would have to be fixed before proceeding.

2. Hydrogen sniffer check

- Once nitrogen integrity was confirmed, a low-pressure hydrogen trial was performed.
- The hydrogen gas sniffer was powered on and allowed to stabilize.
- With a small hydrogen flow enabled and nitrogen still available as a safety purge, the sniffer probe was slowly passed over each connection and around the stack.
- No hydrogen was detected during the trial, confirming good sealing.

3. Electrochemical Leak Detection Test (ELDT)

- After gas-tightness was confirmed externally, an electrochemical leak test was run.
- Hydrogen was supplied at a slightly higher pressure on the anode side while the cathode side remained at lower pressure or inert conditions.
- The cell voltage at open circuit was recorded before and after applying the overpressure.

- The change in OCV (ELDT voltage drop) was approximately 0.008 V, indicating low hydrogen crossover and confirming that there were no significant internal leaks through the membrane at beginning of life.

If any of these tests had indicated a leak, the procedure would have required shutting off hydrogen, venting to nitrogen, disassembling the stack, and checking gasket alignment, MEA placement, and fittings before repeating the checks.

2.5 Test Protocol and Data Logging

Once leak testing was successful, a simple test protocol was implemented to verify that the cell operated correctly.

- **Baseline OCV and polarization**

- Hydrogen was supplied to the anode and air to the cathode.
- The cell was brought to the desired operating temperature using the built-in heaters.
- OCV was recorded after stabilization, and a stepwise current density sweep was performed up to approximately $1.0 \text{ A} \cdot \text{cm}^{-2}$ with short holds at each step.
- Voltage, current, gas flows, pressure, and temperatures were logged by the G40 at a frequency of 1 Hz.

- **Representative cycling**

- Short sequences of OCV holds and mild cycling were used to confirm that the cell behaviour remained stable over several hours.
- Extended accelerated stress tests were not the focus of this report and were only briefly explored.

Log files were exported from FCATS in CSV format for later plotting and inspection.

2.6 Disassembly and Sample Handling

After completing the tests, the Tandem 50 stack was disassembled to inspect the MEA and store the sample.

1. Depressurization and shutdown

- Gas purges and test scripts were stopped in the FCATS software.
- The air valve feeding the air-compressor plate was closed, and the pressure line was disconnected from the plate.
- Hydrogen and nitrogen supplies were shut off at the regulators.

2. Disconnecting instrumentation

- Thermocouples and heater cables were removed from the endplates and air-compressor plate.
- The stack was allowed to cool before handling.

3. Removing the stack from the station

- The TAS (Tandem 50 assembly) was lifted off the G40 platform and placed on the bench.
- Any residual pressure in the stack was relieved via the exhaust valve on the anode plate.

4. Plate-by-plate disassembly

- Bolts and nuts were loosened diagonally and in small increments to avoid sudden release of compression.
- Plates were removed one by one in reverse order of assembly until the polycarbonate spacer and MEA were exposed.
- A photograph of the spacer window was taken to document any signs of mechanical damage or deformation.

5. Sample handling

- The MEA was removed carefully and photographed.

- After cooling, the sample was placed in a labelled plastic bag and stored in a safe, clearly identified location for future inspection or analysis.

This disassembly process not only ensures safe shutdown but also preserves information about how the membrane and spacer have changed during operation.

3. Results

3.1 Assembly and Mechanical Stability

Following the stacking procedure described above, the Tandem 50 assembly remained well aligned, with no visible plate gaps or tilting. The incremental, diagonal tightening sequence successfully prevented bowing of the plates and did not visibly shift the MEA.

When the assembly was transferred onto the G40 station and connected to the gas arms, the stack remained mechanically stable. Minor adjustments were needed to the arm positions and cable routing to avoid introducing unwanted torque or bending, but no re-stacking was required.

3.2 Leak-Test Outcome

The nitrogen leak check showed no soap bubbles at any fittings or connections. During the subsequent low-pressure hydrogen trial, the gas sniffer did not detect hydrogen at any external joints or along the gas arms.

The electrochemical leak detection test produced a small OCV drop of approximately 0.008 V when the hydrogen overpressure was applied. This value is consistent with a low hydrogen crossover rate and indicates that the internal sealing of the MEA and gaskets was successful. No abnormal drops in OCV or sudden jumps in current were observed during this test.

Overall, the leak tests confirm that the assembly and G40 setup procedure produced a gas-tight cell suitable for further electrochemical testing.

3.3 Representative Operating Behaviour

Under hydrogen/air operation at the target temperature, the cell reached a stable OCV in the expected range for a PFSA-based MEA. A simple polarization sweep showed that:

- The voltage decreased smoothly with increasing current density,
- No sudden steps or irregularities occurred during the sweep, and

- The cell maintained stable voltage at a fixed current for several minutes without obvious drift.

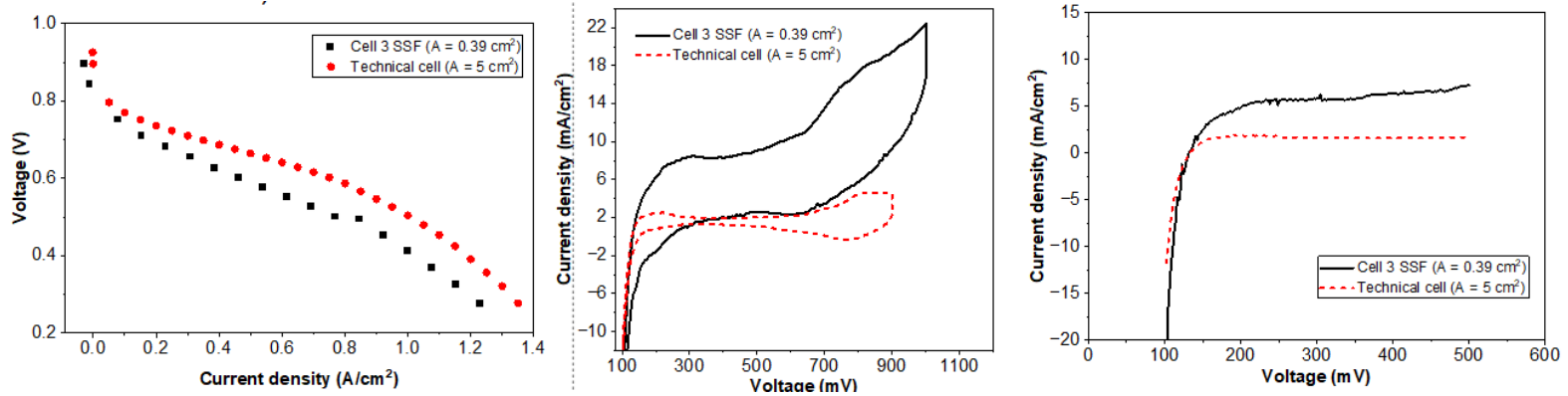


Figure 2 . Comparison of representative electrochemical behavior between the 0.39 cm² small-scale SSF cell (black) and the 5 cm² Tandem 50 research cell (red), including: (a) polarization curves, (b) cyclic current–voltage behavior near open circuit, and (c) low-voltage sweep characteristics.

These observations indicate that the stack assembly, gas distribution, and electrical connections were functioning correctly. Detailed performance analysis and long-duration accelerated stress tests were beyond the scope of this report; the main focus was confirming that the mechanical and leak-tight setup worked as intended.

4. Discussion

4.1 Importance of Correct Stacking and Alignment

The MEA stacking procedure is the most critical hands-on step in preparing the Tandem 50 research cell. The experience from this term showed that:

- Using the cathode endplate as a consistent reference for orientation simplified all later steps.
- Aligning the “red circle dots” on the gold plate and ensuring that all labels were on the same side reduced confusion about anode and cathode routing.
- Centring the MEA visually within the spacer window and tightening bolts diagonally in small increments helped avoid uneven compression and minimized the risk of premature mechanical failure.

These practical details, while simple, can easily be overlooked by new users and are key to obtaining reliable test results.

4.2 Practical Challenges and Solutions in G40 Setup

Working with the G40 station and Tandem 50 hardware highlighted several practical challenges:

- **Arm weight and balance** – The flexible gas arms can apply torque to the stack if they are not supported or angled correctly. Positioning the cell slightly away from the window and adjusting the arm routing mitigated this issue.
- **Thermal insulation** – Exposed red insulation on the arms was identified as a potential heat-loss path. Covering these areas with additional white insulation helped maintain more uniform temperatures.
- **Cable and thermocouple routing** – Wires naturally want to hang down and may rest on hot endplates. Careful routing and tying back of cables were necessary to prevent insulation damage or electrical noise.

By documenting these issues and the steps taken to address them, the procedure becomes more robust and easier to follow for future students.

5. Conclusions and Recommendations

5.1 Conclusions

The work completed this term established a clear, repeatable procedure for assembling and operating a 5 cm² Tandem 50 research single cell on the Greenlight G40 test station. The stacking sequence—using the cathode endplate as the reference, aligning the gold current collector and reaction distribution plates, centring the MEA within the spacer, and tightening diagonally in small increments—produced a mechanically stable, well-aligned assembly.

Integration of the stack with the G40 gas arms, thermocouples, heaters, and pressure line was completed without introducing unwanted torque or thermal weak points. After setup, nitrogen and hydrogen leak checks confirmed gas-tight operation, and the electrochemical leak detection test indicated low hydrogen crossover at beginning of life.

Representative OCV and polarization measurements demonstrated that the assembled cell could be operated reliably under hydrogen/air conditions. The overall workflow—from stacking through leak testing, operation, and disassembly—provides a practical reference for future FCREL users and supports consistent operation of the Tandem 50 hardware.

5.2 Recommendations

Maintain an updated, step-by-step stacking and G40 setup checklist to reduce variability between operators. A short hands-on orientation for new co-op students, using real hardware and annotated photos, would further improve consistency and reduce assembly errors.

During long-duration experiments, incorporate periodic ELDT measurements to track internal leak development. Mechanical supports for the gas arms should also be considered to minimize torque on the stack, especially if the fixture remains on the station for extended periods.

Finally, continue archiving assembly photos, disassembly photos, and MEA condition after testing. Building a reference library of typical compression patterns and failure modes will help diagnose issues more quickly and improve the reliability of future experiments.

6. References

- [1] Lim, C., et al., “Membrane degradation during combined chemical and mechanical accelerated stress testing of polymer electrolyte fuel cells,” *Journal of Power Sources*, vol. 257, pp. 102–110, 2014.
- [2] FCReL Powertech Labs, “G40 Set Up Notes,” internal procedure, 2025.
- [3] Tran, T. N., “Assembly and Testing of a Small-Scale PEM Fuel Cell Using the Greenlight G20 Test Station,” Co-op Technical Report – Work Term 1, Simon Fraser University, 2025.
- [4] Simon Fraser University, School of Mechatronic Systems Engineering, “Technical Report Guidelines,” revised 2024.
- [5] Simon Fraser University, “Assignment #4.2: Technical Report (Co-op Work Term #2),” Canvas, 2025.

AI Usage Statement

AI (ChatGPT by OpenAI) was used to improve grammar, sentence clarity, and overall readability of the report. All technical content, analysis, and conclusions were produced and verified by the author.